

```
import pandas as pd
import numpy as np
```

```
url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv"
df = pd.read_csv(url)

df.head()
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

```
print(df.info())
print(df.describe())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   sepal_length    150 non-null   float64
1   sepal_width     150 non-null   float64
2   petal_length    150 non-null   float64
3   petal_width     150 non-null   float64
4   species         150 non-null   object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
None
```

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333
std	0.828066	0.435866	1.765298	0.762238
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

```
grouped = df.groupby("species")
```

```
summary_stats = grouped.agg(['mean', 'median', 'min', 'max', 'std'])

print(summary_stats)
```

species	sepal_length					std	sepal_width		
	mean	median	min	max	mean		median	min	
setosa	5.006	5.0	4.3	5.8	0.352490	3.428	3.4	2.3	
versicolor	5.936	5.9	4.9	7.0	0.516171	2.770	2.8	2.0	
virginica	6.588	6.5	4.9	7.9	0.635880	2.974	3.0	2.2	

species	max	std	petal_length				std	petal_width
			mean	median	min	max		
setosa	4.4	0.379064	1.462	1.50	1.0	1.9	0.173664	
versicolor	3.4	0.313798	4.260	4.35	3.0	5.1	0.469911	
virginica	3.8	0.322497	5.552	5.55	4.5	6.9	0.551895	

species	median min max std			
	setosa	0.2	0.1	0.6
versicolor	1.3	1.0	1.8	0.197753
virginica	2.0	1.4	2.5	0.274650

```
setosa_list = df[df['species']=='setosa']['sepal_length'].tolist()
versicolor_list = df[df['species']=='versicolor']['sepal_length'].tolist()
virginica_list = df[df['species']=='virginica']['sepal_length'].tolist()

print("Setosa:", setosa_list[:5])
print("Versicolor:", versicolor_list[:5])
print("Virginica:", virginica_list[:5])
```

```
Setosa: [5.1, 4.9, 4.7, 4.6, 5.0]
Versicolor: [7.0, 6.4, 6.9, 5.5, 6.5]
Virginica: [6.3, 5.8, 7.1, 6.3, 6.5]
```

```
setosa = df[df['species'] == 'setosa']
versicolor = df[df['species'] == 'versicolor']
virginica = df[df['species'] == 'virginica']
```

```
print("Statistics for Iris Setosa")
print(setosa.describe())

print("\nStatistics for Iris Versicolor")
print(versicolor.describe())

print("\nStatistics for Iris Virginica")
print(virginica.describe())
```

```
Statistics for Iris Setosa
      sepal_length  sepal_width  petal_length  petal_width
count      50.00000      50.000000      50.000000      50.000000
mean         5.00600         3.428000         1.462000         0.246000
std          0.35249         0.379064         0.173664         0.105386
min           4.30000         2.300000         1.000000         0.100000
25%           4.80000         3.200000         1.400000         0.200000
50%           5.00000         3.400000         1.500000         0.200000
75%           5.20000         3.675000         1.575000         0.300000
max           5.80000         4.400000         1.900000         0.600000
```

```
Statistics for Iris Versicolor
      sepal_length  sepal_width  petal_length  petal_width
count      50.000000      50.000000      50.000000      50.000000
mean         5.936000         2.770000         4.260000         1.326000
std          0.516171         0.313798         0.469911         0.197753
min           4.900000         2.000000         3.000000         1.000000
25%           5.600000         2.525000         4.000000         1.200000
50%           5.900000         2.800000         4.350000         1.300000
75%           6.300000         3.000000         4.600000         1.500000
max           7.000000         3.400000         5.100000         1.800000
```

```
Statistics for Iris Virginica
      sepal_length  sepal_width  petal_length  petal_width
count      50.00000      50.000000      50.000000      50.000000
mean         6.58800         2.974000         5.552000         2.026000
std          0.63588         0.322497         0.551895         0.27465
min           4.90000         2.200000         4.500000         1.40000
25%           6.22500         2.800000         5.100000         1.80000
50%           6.50000         3.000000         5.550000         2.00000
75%           6.90000         3.175000         5.875000         2.30000
max           7.90000         3.800000         6.900000         2.50000
```

```
print("Percentiles for Sepal Length")

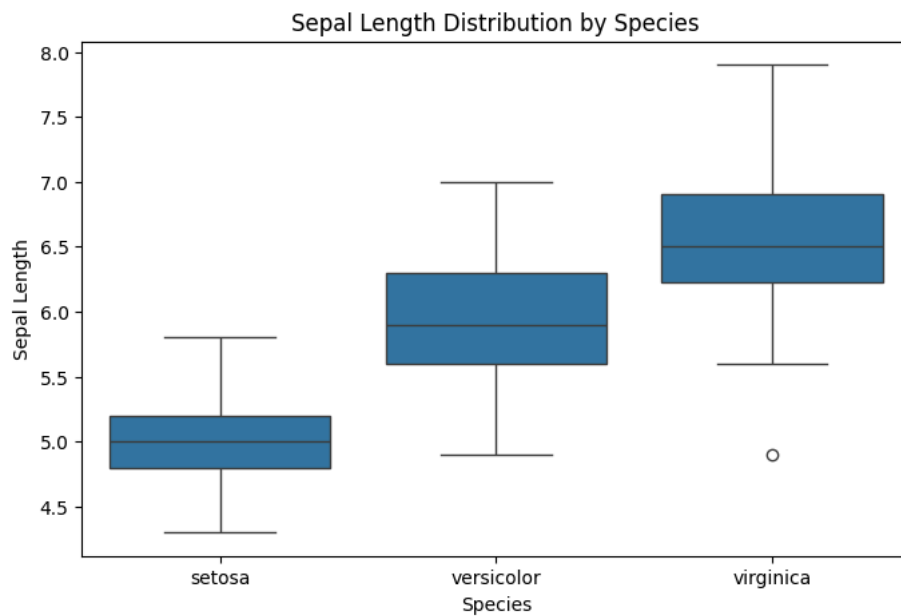
print("25th percentile:", np.percentile(df['sepal_length'],25))
print("50th percentile:", np.percentile(df['sepal_length'],50))
print("75th percentile:", np.percentile(df['sepal_length'],75))
```

```
Percentiles for Sepal Length
25th percentile: 5.1
50th percentile: 5.8
75th percentile: 6.4
```

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

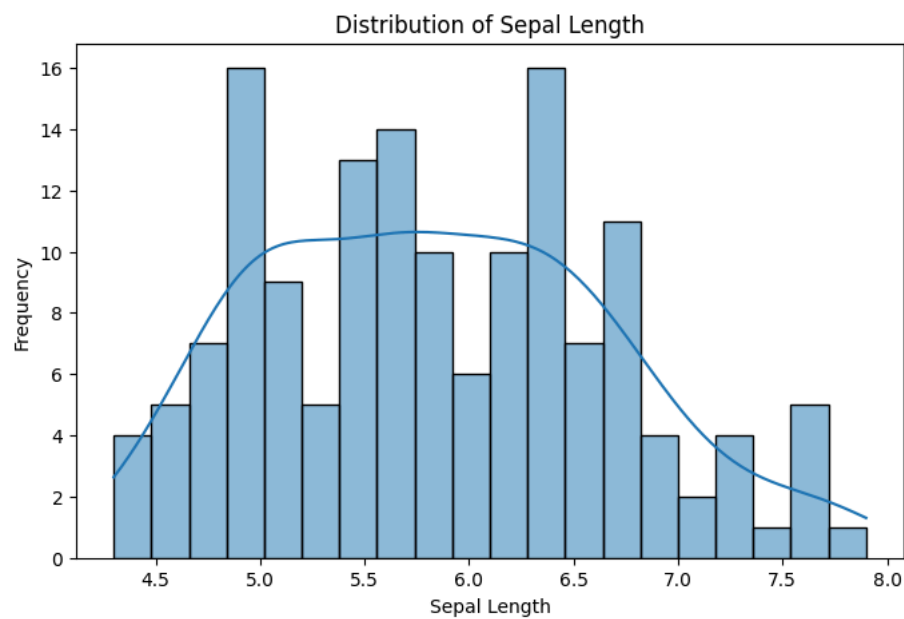
```
plt.figure(figsize=(8,5))
sns.boxplot(x="species", y="sepal_length", data=df)

plt.title("Sepal Length Distribution by Species")
plt.xlabel("Species")
plt.ylabel("Sepal Length")
plt.show()
```



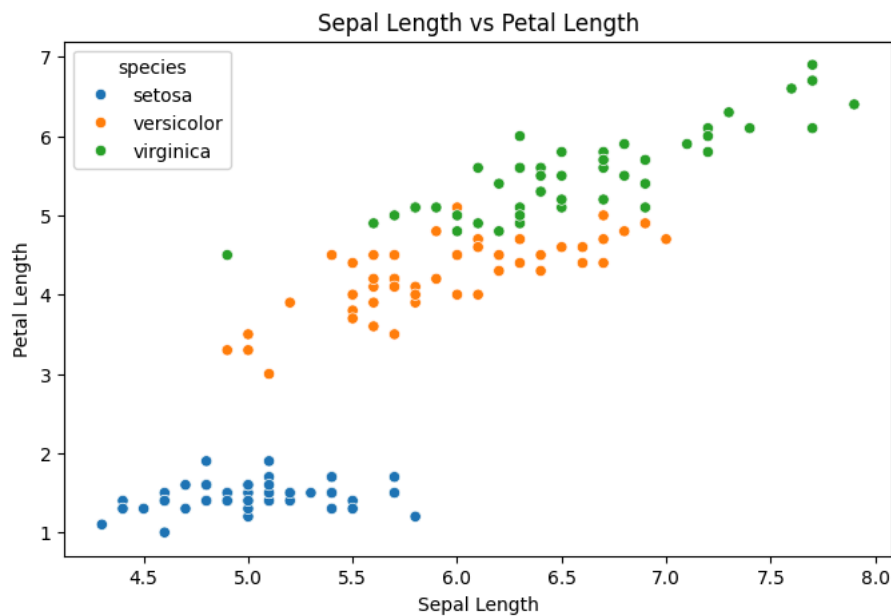
```
plt.figure(figsize=(8,5))
sns.histplot(df["sepal_length"], bins=20, kde=True)
```

```
plt.title("Distribution of Sepal Length")
plt.xlabel("Sepal Length")
plt.ylabel("Frequency")
plt.show()
```

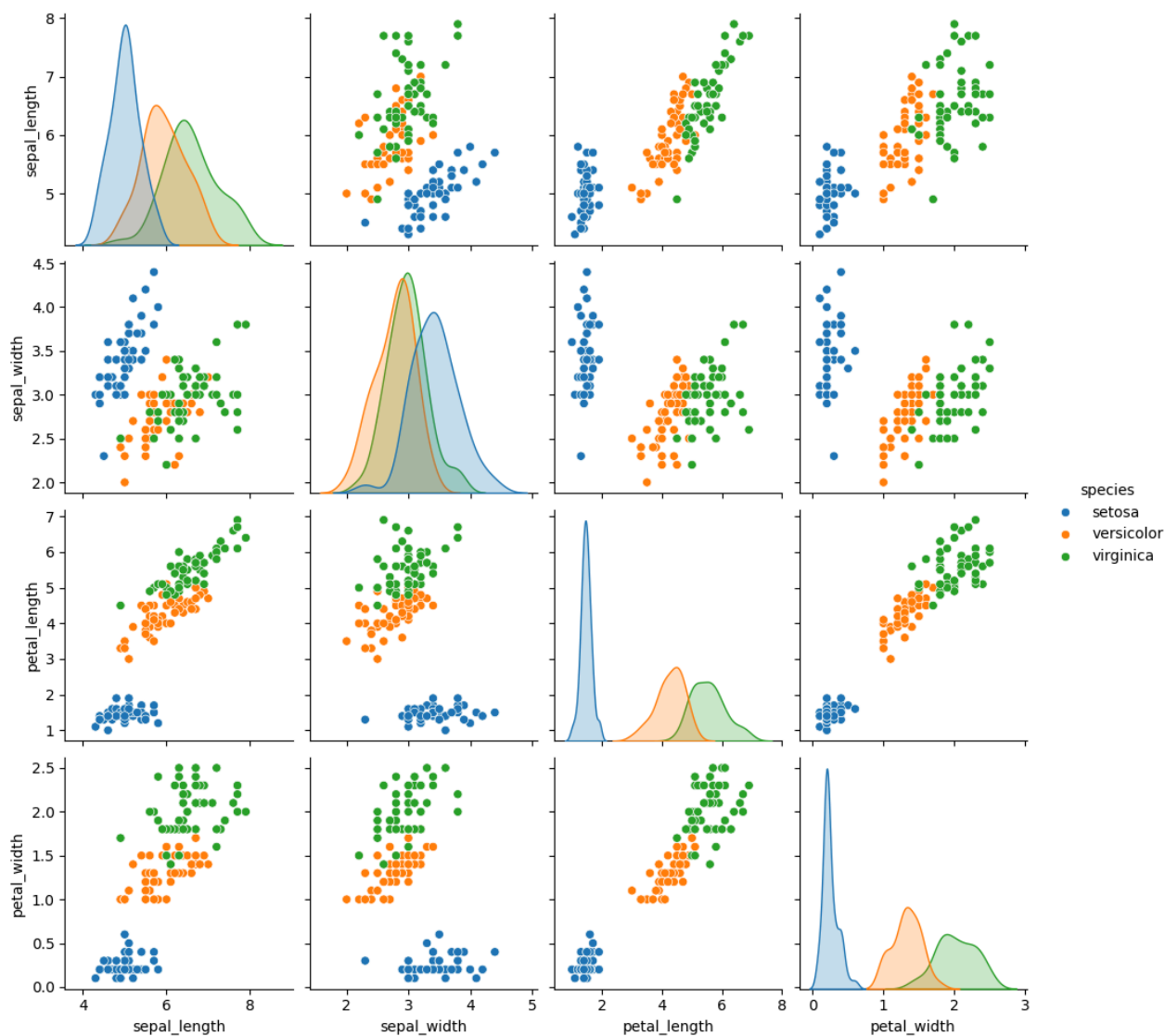


```
plt.figure(figsize=(8,5))
sns.scatterplot(x="sepal_length", y="petal_length", hue="species", data=df)
```

```
plt.title("Sepal Length vs Petal Length")
plt.xlabel("Sepal Length")
plt.ylabel("Petal Length")
plt.show()
```



```
sns.pairplot(df, hue="species")  
plt.show()
```



```
species_mean = df.groupby("species").mean()

species_mean.plot(kind="bar", figsize=(8,5))

plt.title("Mean Measurements by Species")
plt.ylabel("Mean Value")
plt.show()
```

